Experimental Mathematics

Exercises Sheet #1
Error-Correction Codes

Motivation

One of the most useful cryptographic gadget that frequently arises in numerous applications is **Error-Correction Codes** (ECC). While we further give the rigorous definition, the rough purpose of ECC is, as the name suggests, correct errors during transmission through noisy channel or during some complicated processing of data. As previously mentioned, such construction is used in various applications: ranging from error-correction in computer memory or data storage devices (such as HDDs, SSDs, USBs) to fuzzy extractors (recovering strong cryptographic keys from biometrics) or zero-knowledge protocols (namely, zk-STARKs).

Error-Correction Codes are primarily based on linear algebra and polynomials. The primary goal of this exercise sheet is to implement the *Reed-Solomon Codes*, which are among most basic and fundamental types of ECC.

Theory

We define the (n, k) error-correction code $\mathcal{C}$ first. Fix the finite alphabet Σ. You might think of alphabet as the basic building blocks which constitute messages and codewords. Most commonly Σ is just $\{0, 1\}$ (in such case, both messages and codewords are nothing but a sequence of bits) or, for linear codes, some finite field elements $\mathbb{F}_q$ (more on that later).

We call error-correction code $\mathcal{C}$ an (n, k)-code if messages come from set Σ^k, while codewords from Σ^n. Always assume $n \geq k$: we always extend k-sized messages to the larger n-sized codespace. This way, if some errors occur in larger-dimensional space, we hope that we can fix a certain number of errors.

Now we want to induce certain metric over Σ^n. This is done using **Hamming distance**. Let us give the definition.

Definition 0.0.1. The **Hamming distance** $\Delta(\mathbf{x}, \mathbf{y})$ for two codewords $\mathbf{x}, \mathbf{y} \in \Sigma^n$ is defined as the number of positions where $\mathbf{x}$ and $\mathbf{y}$ differ:

$$\Delta(\mathbf{x}, \mathbf{y}) \triangleq \#\{i \in [n] : x_i \neq y_i\}, \quad \mathbf{x}, \mathbf{y} \in \Sigma^n.$$

The Hamming distance naturally induces the norm over Σ^n denoted by $\|\mathbf{x}\|_0 := \Delta(\mathbf{x}, \mathbf{0})$, which is called a **Hamming weight**.